

Q6. Break Affine Cipher

```
C1 = letters.index(c1)
C2 = letters.index(c2)

print("Possible Keys (a, b):\n")

for a in range(1, 26):
    if gcd(a, 26) == 1:
        inv = mod_inverse((P1 - P2) % 26, 26)
        if inv is not None:
            if ((a * ((P1 - P2) % 26)) % 26) == 1:
                b = (C1 - a * P1) % 26
                print("a =", a, " b =", b)
```

... Possible Keys (a, b):

a = 3 b = 15

Q7. Decrypt Simple Substitution Cipher

```
    '-': 'W'
}

print("Decrypted Message:\n")

for ch in cipher:
    if ch in substitution:
        print(substitution[ch], end="")
    else:
        print(ch, end="")
```

... Decrypted Message:

AGEETGHAOONSRIFLNOIEYOIEORFHNSRIFTFDNHOOFAARRVFSRCESFTFG
UFFOASTRINURFFSMNSPRFOEURIFAORASTLCSEURIMANSLUASWIOFDFS
RIHNMLFAORONTFOIEERBUEMRIFHFBRFCFEBRIFTFARIOIFATALFFHNSFB
UEMRIFRUFFRIUEPGIRIFOIERBNBRCBFFREPR

Q8. Keyword Monoalphabetic Cipher

```
# Decryption
decrypted = ""

for ch in encrypted:
    if ch.isalpha():
        index = cipher.index(ch)
        decrypted += alphabet[index]
    else:
        decrypted += ch

print("Decrypted Text :", decrypted)
```

```
... Plain Alphabet : ABCDEFGHIJKLMNOPQRSTUVWXYZ
Cipher Alphabet: CIPHERABDFGJKLMNOPQRSTUVWXYZ
```

```
Enter Plain Text: HELLO
Encrypted Text : BEJJM
Decrypted Text : HELLO
```

Q9. Playfair Cipher Decryption

```
Enter Cipher Text:
KXJEY UREBE ZWEHE WRYTU HEYFS KREHE GOYFI WTTTU OLKSY CAJPO BOTEI ZONTX BYBNT GONEY CUZWR GDSON SXBOU YWRHE BAAHY USEDQ

Decrypted Text:
IZGKCMIMKCI XVFCGUNDLZCFHGTIMKCF FNHGGXSSLZMPITHDXBFVHALKXKMOSZYHYAQKMOGCM LXVNBKBTMOISHAWCZNCFA XBCWL IYT
```

Q10. Playfair Cipher Encryption

```
return cipher

plain = input("Enter Plain Text: ")

print("\nEncrypted Text:")
print(encrypt(plain))
```

```
... Enter Plain Text: hello

Encrypted Text:
MARVAN
```